Conditional closed-loop gradient and finite-difference error bounds

EAGM 2.0 research notes; elementary conditional results

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1 A forward-error non-implication

Set $F(x) = 0$ and $\widehat{F}_\epsilon(x) = \epsilon \sin(x/\epsilon^2)$. Uniform forward error is at most ϵ, but derivative error at zero is $1/\epsilon$. Both maps are smooth. Therefore forward error alone, including a small empirical mean squared error, cannot control Jacobian error without additional uniform regularity assumptions. This is a counterexample, not a diagnosis that the fitted neural networks realize this oscillation.

2 Coordinates and explicit assumptions

Let z include physical, execution, previous-action and reference information sufficient to be a complete state. For fixed discrete execution configuration, define $T_\theta(z) = F(z, \pi_\theta(z))$ and $\widehat{T}_\theta(z) = \widehat{F}(z, \pi_\theta(z))$. Both use the same coordinates and policy. Any latent model application requires an independently justified differentiable representation/lifting; the deployed GRU is not automatically such a representation. Constants below are uniform on a convex neighborhood containing both rollouts and the connecting segments, for the parameter positions considered. Assume $F, \widehat{F}, \pi$ differentiable, with

$$\begin{aligned} \|F_z\|, \left\|\widehat{F}_z\right\| \le L_z, \qquad \|F_a\|, \left\|\widehat{F}_a\right\| \le L_a, \qquad \|\pi_z\| \le P_z, \qquad \|\pi_\theta\| \le P_\theta, \\ \left\|\widehat{F} - F\right\| \le \epsilon_F, \qquad \left\|\widehat{F}_z - F_z\right\| \le \epsilon_z, \qquad \left\|\widehat{F}_a - F_a\right\| \le \epsilon_a. \end{aligned}$$

Use joint distance $\|z - z'\| + \|a - a'\|$; suppose F_z, F_a are Lipschitz with constants K_z, K_a, and π_z, π_θ are Lipschitz in z with constants Q_z, Q_θ. Then valid constants for T_z, T_θ are

$$\begin{aligned} L &= L_z + L_a P_z, & B &= L_a P_\theta, \\ K_A &= (K_z + K_a P_z)(1 + P_z) + L_a Q_z, & K_B &= K_a (1 + P_z) P_\theta + L_a Q_\theta, \\ \epsilon_A &= \epsilon_z + \epsilon_a P_z, & \epsilon_B &= \epsilon_a P_\theta. \end{aligned}$$

Justification. The chain rule gives $T_z = F_z + F_a \pi_z$ and $T_\theta = F_a \pi_\theta$. Bound each product difference by adding and subtracting one common-factor product. The policy changes the joint input distance by at most $1 + P_z$. At the same state the policy factors cancel, giving ϵ_A, ϵ_B. Bounds for both model and true derivatives give the common L.

3 State-estimation and sensitivity errors

Let $e_t = \|\widehat{z}_t - z_t\|$, $S_t = \partial z_t / \partial \theta$, $\widehat{S}_t = \partial \widehat{z}_t / \partial \theta$, $s_t = \|S_t\|$ and $E_t = \left\|\widehat{S}_t - S_t\right\|$. Initial history estimation may have both $e_0 > 0$ and $E_0 > 0$.

Proposition 1 *Under the assumptions above,*

$$e_{t+1} \le L e_t + \epsilon_F, \qquad s_{t+1} \le L s_t + B, \tag{1}$$

$$E_{t+1} \le L E_t + (\epsilon_A + K_A e_t) s_t + \epsilon_B + K_B e_t. \tag{2}$$

Consequently $e_t \le L^t e_0 + \epsilon_F \sum_{i=0}^{t-1} L^i$, $s_t \le L^t s_0 + B \sum_{i=0}^{t-1} L^i$, *and*

$$E_t \le L^t E_0 + \sum_{k=0}^{t-1} L^{t-1-k}\{(\epsilon_A + K_A e_k)s_k + \epsilon_B + K_B e_k\}.$$

Proof. Add and subtract $F(\widehat{z}, \pi(\widehat{z}))$ for the state bound. The chain rule gives $S_{t+1} = A_t S_t + B_t$ and its model counterpart. Subtract them as $\widehat{A}_t(\widehat{S}_t - S_t) + (\widehat{A}_t - A_t)S_t + \widehat{B}_t - B_t$. At their respective states the derivative discrepancies are bounded by $\epsilon_A + K_A e_t$ and $\epsilon_B + K_B e_t$. Apply triangle and operator-norm inequalities, then induction. □

If the saved physical start and observed history are fixed while optimizing theta, both initial sensitivities are zero: $s_0 = E_0 = 0$. This does not set $e_0 = 0$ for hidden-state estimation. If history is itself generated by theta, its derivative cannot be discarded. The implementation's frozen starts analyze a conditional rollout objective, not derivatives of its data-collection distribution.

4 Reward and policy regularity

Write the per-step scalar return as $R(z, a)$ and its modeled counterpart $\widehat{R}(z, a)$; these may absorb next-state rewards through F and previous action through z. Assume $\left\|\widehat{R}_z\right\| \le R_z^*$, $\left\|\widehat{R}_a\right\| \le R_a^*$ and the same true bounds, with same-input derivative errors $\epsilon_{Rz}, \epsilon_{Ra}$. Suppose true R_z, R_a have joint Lipschitz constants K_{Rz}, K_{Ra}. For closed-loop $r_\theta(z) = R(z, \pi_\theta(z))$, define

$$C = R_z^* + R_a^* P_z, \qquad \epsilon_C = \epsilon_{Rz} + \epsilon_{Ra} P_z, \qquad \epsilon_D = \epsilon_{Ra} P_\theta,$$
$$K_C = (K_{Rz} + K_{Ra} P_z)(1 + P_z) + R_a^* Q_z, \qquad K_D = K_{Ra}(1 + P_z)P_\theta + R_a^* Q_\theta.$$

For $J = \sum_{t=0}^{H-1} \gamma^t r_\theta(z_t)$ and $0 \le \gamma \le 1$,

$$\left\|\nabla \widehat{J} - \nabla J\right\| \le \sum_{t=0}^{H-1} \gamma^t \left[C E_t + (\epsilon_C + K_C e_t)s_t + \epsilon_D + K_D e_t\right]. \tag{3}$$

Proof. The contribution is $C_t S_t + D_t$ with $C_t = \partial_z r_\theta$, $D_t = \partial_\theta r_\theta$. Subtract modeled and true contributions, adding and subtracting $\widehat{C}_t S_t$. The reward product bounds follow identically to the transition ones; sum the triangle inequalities. □

For a learned visual reward, errors $\epsilon_{Rz}, \epsilon_{Ra}$ are not zero by definition. Even for an analytic physical reward, composing it with an imperfect transition creates reward derivative discrepancy. Changing the start distribution adds a separate gradient-distribution error. The bound can grow exponentially when $L > 1$; no measured or certified constants are supplied, so it is not a numerical performance guarantee.

5 Finite differences matching implemented stencils

For coordinate restriction $f(s) = J(\theta + se_j)$ assume $f \in C^3$ on the full feasible stencil interval, $|f'''| \le M_3$, and each reported objective has error at most η. These are deterministic local conditions, not inferred from scale agreement.

Proposition 2 *For symmetric central differences at* $\pm h$*, error is at most* $M_3 h^2/6 + \eta/h$*. For unequal offsets* $-a, 0, b$ *with* $a, b > 0$*, weights*

$$w_- = -\frac{b}{a(a+b)}, \quad w_0 = \frac{b-a}{ab}, \quad w_+ = \frac{a}{b(a+b)}$$

give error at most $M_3 ab/6 + \eta(|w_-| + |w_0| + |w_+|)$*. At the exact lower boundary the stencil* $(-3f(0) + 4f(h) - f(2h))/(2h)$ *has error at most* $M_3 h^2/3 + 4\eta/h$*; reflection gives the upper-boundary result.*

Proof. Taylor expansion cancels constant and quadratic terms, while the weighted linear term equals $f'(0)$. For the unequal case the third-order remainder bound is $M_3[ba^2 + ab^2]/[6(a+b)] = M_3ab/6$. The symmetric case follows by $a = b = h$. For the forward case use the integral Taylor remainder. Its kernel is $[4(h-s)^2 - (2h-s)^2]/(4h)$ for $0 \le s \le h$ and $-(2h-s)^2/(4h)$ for $h \le s \le 2h$. Both are nonpositive and their total integral is $-h^2/3$. Noise contributes the sum of absolute weights times η. $\square$

The implementation reduces h if needed to keep both one-sided points feasible, and includes cached center queries in accounting. Clipping a symmetric stencil without correcting its weights does not satisfy this result. At a contact or clipping kink, C^3 fails and multi-scale agreement cannot repair the missing hypothesis. With stochastic rollouts an almost-sure bounded-error or separate probabilistic noise model is needed; these inequalities alone give no confidence interval.

6 Actual policy update, not raw-gradient rhetoric

If the evaluation objective has L_J-Lipschitz gradient on the actual update segment and $\|\widehat{g} - \nabla J\| \le \delta$, then for any feasible actual step d,

$$J(\theta + d) - J(\theta) \ge \widehat{g}^T d - \delta \|d\| - \frac{L_J}{2} \|d\|^2 .$$

Integrate the gradient along the segment and apply Cauchy–Schwarz to prove this inequality. Adam and projection require their actual resulting d; a high cosine alone neither supplies δ nor a safe step size. No convergence, improvement, contact differentiability, latent sufficiency or novelty claim follows without these additional assumptions.